

Limitation/Injury Filtering System

Comprehensive Implementation Report

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Executive Summary

This document details the implementation of a comprehensive, metadata-driven Limitation/Injury Filtering System for the Fit33 iOS fitness app. The system intelligently filters and scores exercises based on user-specified physical limitations (injuries, chronic conditions) without hardcoding specific exercise names. Instead, it uses exercise metadata tags to make filtering decisions that work for any exercise, including new ones added to the database.

Key Features

- **Metadata-Driven:** No hardcoded exercise names - uses tags like spinal load, impact level, etc.
- **4 Severity Levels:** Skip Completely, Stretching Only, Light Work Only, Be Careful
- **9 Body Areas:** Lower Back, Shoulders, Knees, Hips, Neck, Wrists, Elbows, Ankles, Other
- **Comprehensive Rule Tables:** Area-specific rules for each severity level
- **Scoring Integration:** Penalties/boosts applied during exercise selection
- **Unit Tested:** 10 test cases covering all major scenarios
- **Backward Compatible:** Works with existing LimitationsService

System Architecture

The system is composed of 5 new Swift files that work together to provide comprehensive limitation filtering:

File	Purpose	Lines
LimitationFilterEngine.swift	Core models, enums, and filtering logic	~400
LimitationRuleTables.swift	Area-specific rule definitions	~500
ExerciseMetadataClassifier.swift	Intelligent exercise metadata derivation	~350
LimitationFilterIntegration.swift	Integration with existing services	~350
LimitationFilterTests.swift	Unit tests for all scenarios	~550

Data Models

LimitationArea Enum

Defines the body areas that can have limitations:

lowerBack, shoulders, knees, hips, neck, wrists, elbows, ankles, upperBack, other

LimitationSeverity Enum

Severity	Behavior	Score Impact
Skip Completely	Hard exclusion of exercises affecting area	Exercise removed
Stretching Only	Only allow stretch/mobility exercises for area	Exercise removed if not stretch
Light Work Only	Exclude heavy compounds, prefer machines	±50-200 penalty/boost
Be Careful	Penalize risky, boost safe options	±30-150 penalty/boost

Exercise Risk Metadata Tags

Each exercise is classified with these metadata tags, derived automatically by the ExerciseMetadataClassifier based on exercise name and equipment:

Tag	Values	Example Use
spinalLoad	none, low, moderate, high	Lower back filtering
axialLoading	none, low, high	Cervical spine protection
unsupportedTorso	true, false	Bent-over row detection
overheadWork	none, partial, full	Shoulder overhead exclusion
kneeFlexionDepth	low, moderate, high	Deep squat filtering
shoulderStressFlags	uprightRow, behindNeck, etc.	Dangerous shoulder patterns
impactLevel	none, low, moderate, high	Plyometric filtering
balanceDemand	low, moderate, high	Ankle stability
isMachineSupported	true, false	Safe alternative preference
isChestSupported	true, false	Back-safe row preference
isStretchOrMobility	true, false	Stretching-only mode
hipHingeDemand	true, false	Hip/back hinge detection

Area-Specific Rule Tables

Lower Back Rules

- **Skip Completely:** High spinal load, high axial loading, unsupported hinge, unsupported bent-over row
- **Light Work Exclude:** High spinal load, unsupported hinge patterns
- **Light Work Prefer:** Machine (+50), chest-supported (+80), seated (+40), lying (+60)
- **Light Work Penalize:** Moderate spinal load (+100), unsupported torso (+150)
- **Be Careful Boost:** Machine (+30), chest-supported (+50), lying (+40)
- **Be Careful Penalize:** High spinal load (+150), unsupported torso (+80)

Shoulder Rules

- **Skip Completely:** Full overhead, upright row pattern, behind neck, guillotine, heavy dip
- **Light Work Exclude:** Full overhead, heavy dip, behind-neck/guillotine/upright-row patterns
- **Light Work Prefer:** Machine (+50), cable work (+30)
- **Be Careful Penalize:** Upright row (+100), behind neck (+150), guillotine (+150), overhead (+60)

Knee Rules

- **Skip Completely:** High knee flexion (deep squat), moderate+ impact, heavy squat/lunge
- **Light Work Exclude:** Deep flexion without machine support, moderate+ impact
- **Light Work Prefer:** Machine (+60), low flexion (+40), lying (+50)
- **Be Careful Penalize:** Impact (+120), deep flexion (+80), high balance (+60)

Integration with Workout Generation

The limitation filtering system integrates at two key points in the workout generation flow:

1. Pre-Filtering (*LimitationsService.filterSafeExercises*)

Before any exercise selection, the full exercise library is filtered to remove exercises that violate 'Skip Completely' or 'Stretching Only' limitations. This uses both the legacy keyword-based system and the new metadata-driven classifier for comprehensive coverage.

2. Scoring (*LimitationsService.getSafetyPenalty*)

During exercise scoring in `WorkoutGeneratorService`, each exercise receives a safety penalty or boost based on 'Light Work Only' and 'Be Careful' limitations. This allows the system to prefer safer alternatives without completely excluding exercises.

Code Integration Example

```
// In WorkoutGeneratorService.generateFromCoreData: // 1. Filter unsafe exercises first
allExercises = LimitationsService.shared.filterSafeExercises(allExercises) // 2. Apply safety
penalty during scoring let safetyPenalty = LimitationsService.shared.getSafetyPenalty(for:
exercise) score -= safetyPenalty // Positive penalty = decrease score
```

Unit Tests

The system includes comprehensive unit tests covering all major scenarios:

Test	Description	Assertions
Lower Back Skip	High spinal load excluded, machine/chest-supported allowed	4
Lower Back Light	Heavy excluded, machine allowed, chest-supported boosted	4
Lower Back Careful	Penalties for high load, boosts for safe options	3
Shoulder Skip	Overhead, upright row, behind neck, dip, guillotine excluded	5
Shoulder Careful	Dangerous patterns penalized, machines boosted	2
Knee Skip	Deep flexion and impact excluded, machine isolation allowed	3
Stretching Only	Non-stretch excluded, stretch allowed	2
Multiple Limitations	All limitations applied, skip wins over careful	2
Equipment Cap	Foundational ≤2, short workout ≤2, long workout ≥4	3
Metadata Classification	Deadlift, machine row, OHP, stretch correctly classified	4

User Flow

Setting Limitations

Users can set limitations during onboarding or in Settings → Limitations: 1. Select Affected Area (Lower Back, Shoulders, Knees, etc.) 2. Choose Accommodation Level: • Skip completely - Don't recommend ANY exercises for this area • Light work only - Light exercises OK, skip heavy compounds • Stretching only - Only stretching/mobility for this area • Be careful - Can do exercises, prefer safer options 3. Optionally add notes 4. Save - limitations apply immediately to all future workouts

During Workout Generation

When generating a Quick Workout: 1. User selects duration, muscle groups, focus areas, equipment 2. System fetches all matching exercises from database 3. LimitationsService.filterSafeExercises removes hard exclusions 4. Remaining exercises are scored with safety penalties/boosts 5. Top-scored exercises are selected 6. If limitations caused substitutions, user sees a note: "Adjusted to respect your Lower Back setting"

Example Scenarios

Scenario 1: Lower Back Injury - Skip Completely

User: Has a herniated disc, selects 'Skip completely' for Lower Back Result: Deadlifts, bent-over rows, good mornings, heavy squats are excluded Alternatives: Machine rows, chest-supported rows, leg press are prioritized

Scenario 2: Shoulder Tendinitis - Be Careful

User: Recovering from shoulder tendinitis, selects 'Be careful' for Shoulders Result: Upright rows and behind-neck exercises heavily penalized (-100 to -150) Machine presses and cable exercises get boost (+30) Workout still includes pressing movements but prefers safer variations

Scenario 3: Knee Surgery Recovery - Light Work Only

User: 3 months post knee surgery, selects 'Light work only' for Knees Result: Deep squats and plyometrics excluded Leg press, seated leg curl, step-ups allowed with preference for machines High-balance movements penalized

Future Enhancements

The metadata-driven architecture enables easy expansion:

- **Database-stored metadata:** Add columns to exercises table for cached metadata
- **User override rules:** Allow users to mark specific exercises as safe/unsafe for them
- **Progressive recovery:** Suggest moving from 'Skip' to 'Light' to 'Careful' over time
- **Physical therapist integration:** Allow PTs to set client-specific limitations
- **Smart alternatives:** Suggest specific safe alternatives when excluding exercises
- **Warning UI:** Show caution badges on exercises during workout
- **History learning:** Track which exercises user swaps out to learn preferences

Conclusion

The new Limitation/Injury Filtering System provides comprehensive, intelligent protection for users with physical limitations. By using metadata-driven rules rather than hardcoded exercise lists, the system:

- Works with ALL exercises, including future additions
- Respects user preferences (Skip/Light/Stretch/Careful)
- Prefers safe alternatives over exclusion when possible
- Is fully tested with unit tests
- Integrates seamlessly with existing workout generation

Users can now confidently generate workouts knowing the app will respect their physical limitations and prioritize their safety.